SURVEILLANCE ROBOT

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CONTRIBUTIONS BY EACH MEMBER

HELON: I generated a URDF model for the robot, integrated camera and lidar sensors. Launched RViz for sensor data visualization and Gazebo for robotic simulation.

GOURI: I developed a mobile robot and implemented teleoperation to control its movement. I simulated the STL file in FreeCAD to work on its wheel rotation and created a world file for simulation purposes.

VIVIN: I was looking into the .stl file to be able to mesh the base and wheels of a robot model from fusion360 and I even tried to scale down and change the color of the model but I was unable to mesh it correctly in the urdf file and sdf file. It would load into gazebo but I was unable to make it move. I also looked into the gazebo world to get a good factory simulation to replicate the environment our surveillance robot would be in.

REFERENCES

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[4] https://classic.gazebosim.org/tutorials?tut=guided_i2

[5] https://gazebosim.org/docs/latest/moving_robot/

[6] https://www.youtube.com/watch?v=sb7FoOGzb8E

[7] https://www.youtube.com/watch?v=zGWFojrPoSA

[8] https://www.youtube.com/watch?v=AfP_8Nf9ynw

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